Computer Graphics BMI007

Prof. Dr. Renato Pajarola
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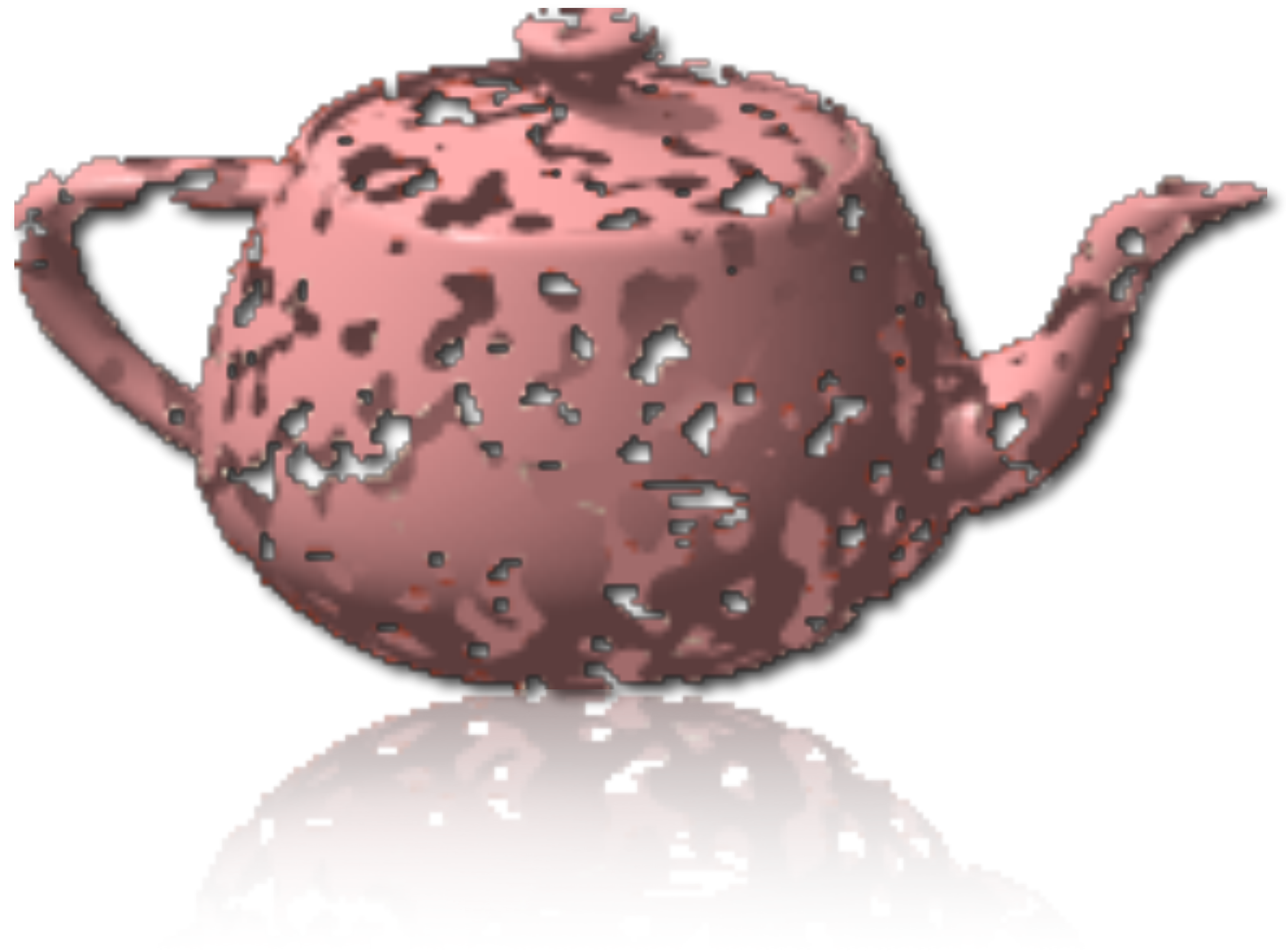
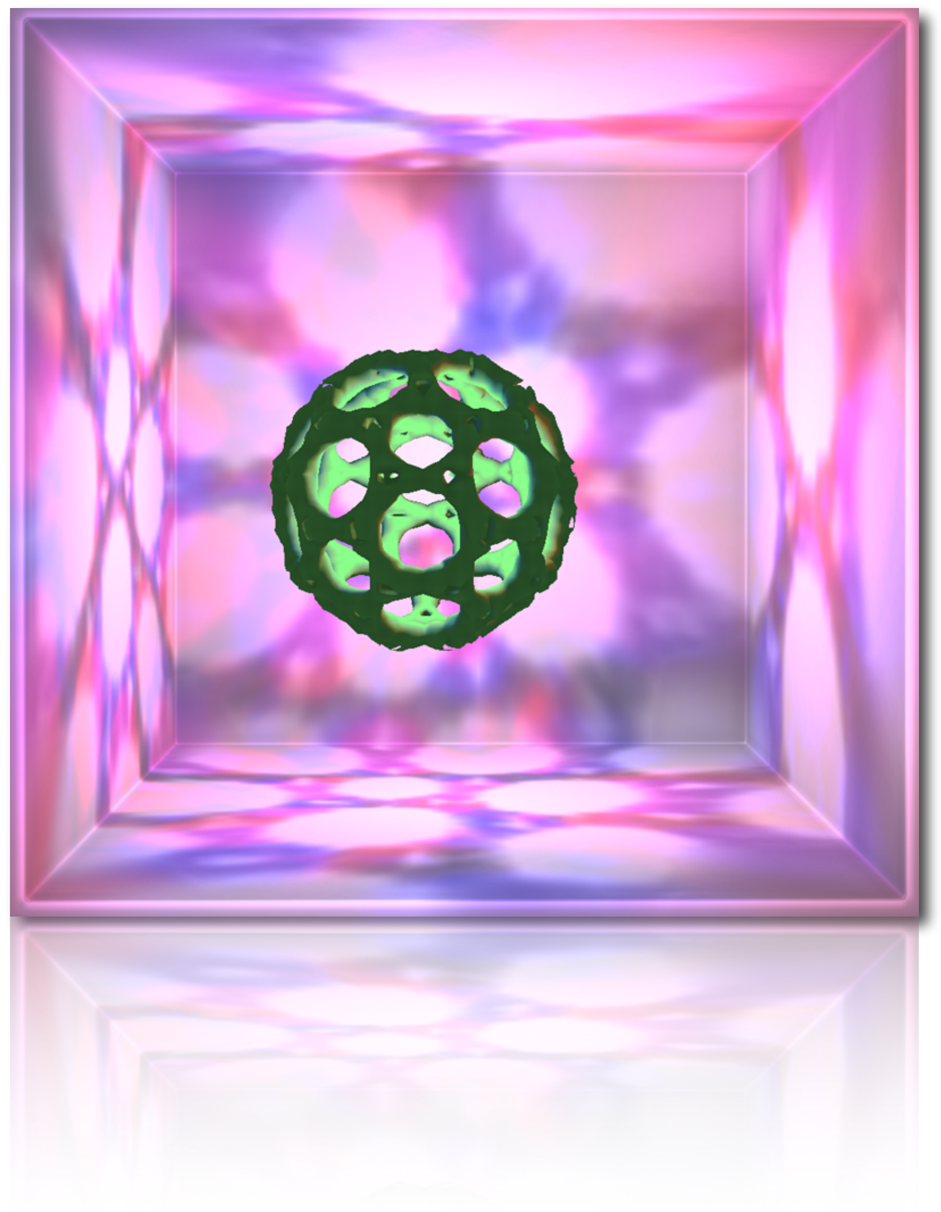




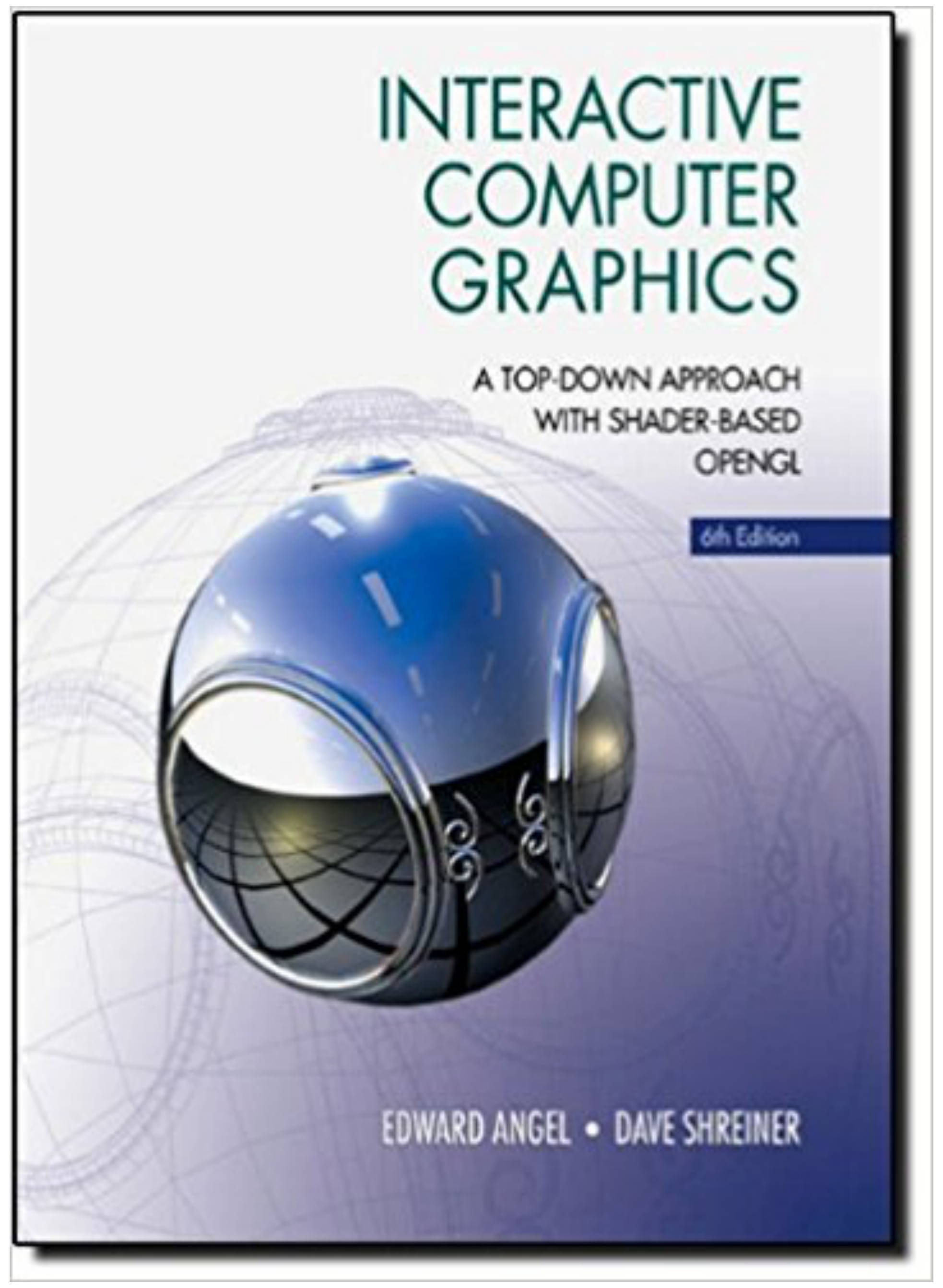
Organization

- Time: Monday 14-15:45 and Thursday 10:15-12:00
- Location: BIN 2.A.10
- Instructor: Prof. Dr. Renato Pajarola
    - Office: BIN 2.C.02
    - Email: pajarola@ifi.uzh.ch
- Teaching assistants:

  Shidong Wang - shwang@ifi.uzh.ch
    - Office: 2.C.05

  Zehang Qiu - zehang@ifi.uzh.ch
    - Office: 2.C.05

Literature



- Required textbook:
  - E. Angel and D. Shreiner. **Interactive Computer Graphics: A top-down approach with Shader-Based OpenGL**. Addison-Wesley.
    - many variations of the book exist with newer variants (e.g. WebGL)
    - available in various forms, used, as rental or ebook
- Secondary literature and programming books:
  - other 3D computer graphics textbooks (using shader based OpenGL)
  - Hughes et al. **Computer Graphics: Principles and Practice**. Addison-Wesley.
  - Pulli et al. **Mobile 3D Graphics with OpenGL ES and M3G**. Morgan Kaufmann.
  - Munshi et al. **OpenGL ES Programming Guide**. Addison-Wesley.
  - E. Angel. **OpenGL: A Primer**. Addison-Wesley.
  - Woo et al. **OpenGL Programming Guide** (Fourth Edition). Addison-Wesley

Links

- OLAT course: https://lms.uzh.ch/url/RepositoryEntry/17827989313
  - lecture slides, examples, announcements and exercises
- Course catalogue: https://studentservices.uzh.ch/uzh/anonym/vvz/index.html#/details/2025/004/E/51311847
  - description, requirements, regulation, lecture and final exam times and locations,
- Class web page: https://www.ifi.uzh.ch/en/vmml/teaching/lectures/computer-graphics-fs26.html
  - general and other additional information
- Further helpful links:
  - Op OpenGL - http://www.opengl.org
  - GLFW - http://www.glfw.org
  - OpenGL programming tutorial - http://nehe.gamedev.net/
  - Visualization and MultiMedia Lab - http://vmml.ifi.uzh.ch/

Exercises

- Simple OpenGL programming exercises accompanying lecture
  - C++ based OpenGL examples that have to be completed and/or extended
- Get to know how a basic OpenGL program looks & feels like
  - also using simple shaders
- Understand the basic representation, processing and rendering of 3D objects
- Exercise Q & A sessions
  - discuss open questions, prepare and review exercises
- Time: Thursday 16:15 - 17:00
- Location: BIN 2.A.01

Exercise Schedule

- Tentative schedule:

Handout	Submission	Topic
Mon, 16 Feb 2026	Mon, 23 Feb 2026	Compile and run OpenGL “Hello Cube”
Mon, 23 Feb 2026	Fri, 6 Mar 2026	3D Geometry
Mon, 9 Mar 2026	Fri, 20 Mar 2026	Transformations
Mon, 23 Mar 2026	Fri, 10 Apr 2026	Illumination
Mon, 13 Apr 2026	Fri, 1 May 2026	Texturing
Mon, 4 May 2026	Fri, 22 May 2026	Ambient occlusion and shadows

- Submission before the start of the next exercise, unless specified otherwise
  - officially binding information and deadlines on OLAT

Exercise Scores

- The exercises are evaluated as follows:
  - Exercise 0: Compile and run OpenGL “Hello Cube” — all pass
    - simple initial code setup and preparation task
  - **Exercise 1: 3D Geometry — 3+1 points**
  - **Exercise 2: Transformations — 3+1 points**
  - **Exercise 3: Illumination — 4+1 points**
  - **Exercise 4: Texturing — 5+1 points**
  - **Exercise 5: Ambient occlusion and shadows — 5+1 points**
- To successfully pass the exercises, at least 10 points have to be achieved
  - 10 is required out of a maximum total of 20 (+5 optional)
  - 11 and above will give up to 10 bonus points (maximum capped)
  - bonus points will be added to the points scored on the final exam

Final Exam

- In addition to passing the exercises, the successful completion of a written final exam is required
  - not achieving the minimal exercises score leads to failing the module
- The final exam will be on Monday June 8th 2026 at 14:00
  - see official course catalogue information for the legally binding information
- The final course grade will be determined by the score from the written final exam
  - plus up to 10 bonus points added from the exercises

This lecture is not about ...

- ... how to design 2D or 3D graphics
- ... how to create computer animations
- ... how to use Photoshop or Illustrator
- ... how to develop in HTML5 or Flash
- ... how to design animated web pages
- ... how to use different graphics APIs/libraries

This lecture is about ...

- ... fundamental concepts and algorithms of 3D graphics
  - mostly for interactive systems such as 3D games, virtual reality environments, scientific visualization
- ... real-time image generation from 3D scenes
- ... geometric transformations and projections
- ... illumination and shading
- ... algorithms and data structures